

14 The ionic product of water, K_w , is affected by temperature.

temperature / °C	$K_w \times 10^{-14} / \text{mol}^2 \text{dm}^{-6}$
10	0.293
40	2.92

Which statement describes what happens as the temperature of water is increased from 10 °C to 40 °C?

- A** pH of water decreases and $[\text{H}^+] = [\text{OH}^-]$
- B** pH of water decreases and $[\text{H}^+]$ is greater than $[\text{OH}^-]$
- C** pH of water increases and $[\text{H}^+] = [\text{OH}^-]$
- D** pH of water increases and $[\text{H}^+]$ is less than $[\text{OH}^-]$